

QoE-Driven Resource Allocation for Stacked Intelligent Metasurface Systems

Hosein Zarini^{*}, S. Mohsen Kazemi^{*}, Jiancheng An^{**}, Ali Movaghar[†], Mehdi Sookhak^{*} and Nuri Yilmazer[§]

^{**}Nanyang Technological University, Singapore

^{*}Texas A&M University, Corpus Christi, TX, USA

[§]Texas A&M University-Kingsville, Kingsville, TX, USA

[†]The University of Michigan, Ann Arbor, MI, USA

Abstract—This endeavor centers around downlink transmission of a base station (BS), outfitted with a stacked intelligent metasurface (SIM) that realizes an energy-efficient wave-domain multi-user beamforming. The underlying network includes mobile devices with multimedia service requirements, including web surfing, HTTP live video streaming and voice-over-LTE (VoLTE). Rather than conventional quality-of-service (QoS) metrics, we assess the satisfaction level of users relying on quality-of-experience (QoE) criteria. Invoking mean opinion score (MOS) as the subjective measurement of QoE, we also evaluate the overall efficacy of this system by posing a resource allocation optimization problem aimed at maximizing the achievable MOS of all users, while adhering to their minimum MOS requirements and the maximum transmit power budget of the BS. Due to the interdependency of optimization variables and non-convex nature of the problem, we first reformulate it in Markov decision process, which captures its dynamic traits. Relying on the MDP model, a conservative Q-learning (CQL) agent is trained for jointly designing the optimization variables, including the BS downlink transmit power, as well as the electromagnetic response of the SIM. In light of real-time mobility of users and the resulting non-trivial network dynamism, we further utilize meta-learning technique to enhance the adaptability and generalization of the CQL agent. Numerically, it is demonstrated that, respectively, 31%, 44% and 26% superior average MOS is achieved, for web, video and audio services, compared to traditional QoS-driven resource allocation.

Index Terms—Conservative Q-learning (CQL), mean opinion score (MOS), meta-learning, quality-of-experience (QoE), stacked intelligent metasurface (SIM).

I. INTRODUCTION

Toward accommodating the ever-growing number of ubiquitously dispersed devices with diverse service requirements, such as pervasive 3D multimedia, HDTV, augmented reality, VoIP, gaming, and e-Health, traditional wireless cellular networks are evolving day in day out [1]. Instead of conventional quality-of-service (QoS) metrics, such as end-to-end latency, data rate, and energy efficiency, multimedia applications often demand a more nuanced evaluation criterion. To this end, quality-of-experience (QoE) as a more generic and personalized criterion is predominantly preferred for analyzing the multimedia service requirements [2]. This criterion is inherently a subjective and user-centered standard that quantifies multimedia quality by directly integrating end-user satisfaction [3]. To effectively exhibit the user-centric level of satisfaction, a specialized grading mechanism is employed which scores

web-based services within a range from 1 to 5, while video and audio use a refined scale from 1 to 4.5 [4]. This narrower grading scope allows for a more precise assessment of user contentment, contrasting with the broader, less subjective range used in QoS evaluations such as bits per second. To formalize multimedia service quality within this framework, objective QoE metrics, such as the mean opinion score (MOS) [5], have been introduced, often expressed as a function of data rate to align technical and perceptual quality indicators. Accommodating diverse multimedia service requirements, however, induces a non-trivial energy consumption to existing infrastructure [6]. Hence, further research on green communication technologies is already ongoing for multimedia applications.

An innovative progression in this way is the concept of stacked intelligent metasurfaces (SIMs), introduced through layering of multiple transmissive metasurfaces, as detailed in [7]. This configuration enables signal processing functions, such as multi-user precoding, to occur directly in the electromagnetic (EM) wave domain, resulting in a reduced demand for radio-frequency chains at transceivers. Consequently, this setup achieves substantial energy savings while operating near-instantaneous (at the speed of light). Thus far, this technology has seen some application in wireless systems, such as multi-user beamforming [8], direction-of-arrival estimation [9], channel estimation [10], and low-earth orbit satellite communications [11]. Thanks to the aforementioned advantages, SIM-enabled base stations (BSs) as a green infrastructure can effectively serve for diverse energy intensive services especially with multimedia requirements—an integration yet to be explored. In summary, *a review of the literature reveals that despite the considerable energy efficiency, SIMs contributes to existing wireless systems, they have never been exploited to support multimedia services.*

This study seeks to bridge an existing research gap in this area by analyzing the efficiency of a wireless network, where a BS employs a SIM to execute wave-domain multi-user beamforming for serving multimedia users. Ensuring optimal system performance in such a setup necessitates a coordination of network resources (such as wave-domain beamforming at the SIM and transmit power of the BS), a task that traditional iterative convex optimization techniques, like those presented in [10], [11], often require complex matrix calculations to achieve and struggle to perform due to their computational

intensity and time demands. Future wireless systems are expected to operate in highly dynamic environments, which require real-time resource management approaches capable of continuously adapting to network dynamism—the core objective that motivated us towards this research. In response to these challenges, this paper introduces the following primary contributions.

- This research investigates a SIM-equipped BS capable of executing EM wave-domain beamforming, specifically designed to serve users with diverse multimedia needs, including web browsing, HTTP-based video streaming, and VoLTE services. To optimize the performance of this system, a comprehensive resource allocation model is developed with the goal of maximizing the cumulative MOS of multimedia users. This is achieved while adhering to each user's minimum MOS requirement and maintaining the BS's maximum transmission power limit. The complexity of this model dominantly stems from the simultaneous and intertwined optimization of variables, including the BS's downlink transmission power and the EM characteristics of the SIM.
- To ensure real-time adaptability to the network's inherent dynamics, a rapid-response resource allocation framework is proposed. Within this framework, the complex, interdependent, and non-convex resource allocation problem is first reformulated as a Markov decision process (MDP) that captures its dynamic features effectively. On this basis, a conservative Q-learning (CQL) agent is trained to jointly regulate the optimization variables. To accommodate remarkable user mobility and significant network variations, the CQL agent is further enhanced with meta-learning, enabling it to quickly adjust to evolving wireless environmental conditions.
- Numerical results from our analysis demonstrate that a SIM-enabled BS serving multimedia applications based on QoE metric, can respectively yield average MOS improvements of 31%, 44% and 26% for web, video and audio services, compared to traditional QoS-based resource allocation.

II. SYSTEM SETUP AND PROBLEM STATEMENT

A. Network Design

As depicted in Fig. 1, we begin with downlink transmission of a BS, outfitted with a SIM, transmitting signals to a set $\mathcal{U} = \{1, 2, \dots, U\}$ of U single-antenna users. Through an antenna array, the BS dispatches U concurrent data streams, each targeted to an individual user. The time is divided into N slots, each with a fixed duration τ .

B. SIM Configuration

As a radome encasing the BS, the SIM consists of a sequence $\mathcal{L} = \{1, 2, \dots, L\}$ comprising L uniformly spaced metasurface layers. Each layer adopts a square uniform planar array (UPA) configuration, encompassing a set $\mathcal{F} = \{1, 2, \dots, F\}$ where $F \geq U$, representing the array of meta-atoms. Integrated within the SIM, an intelligent controller

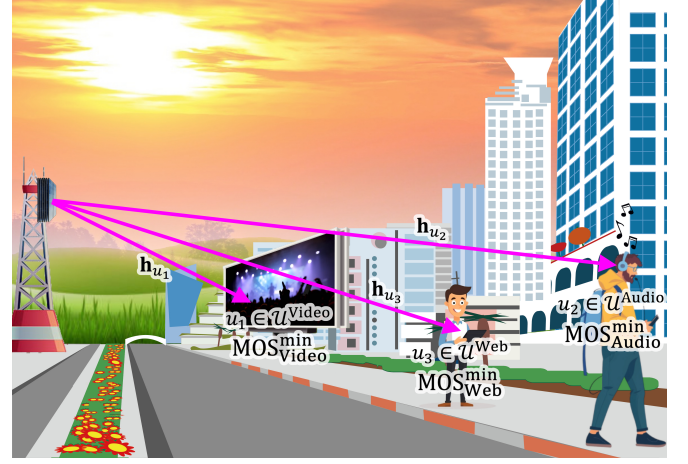


Fig. 1: A schematic of the system.

autonomously adjusts the phase shifts of EM waves as they traverse each meta-atom, enabling the SIM to perform downlink beamforming directly at the EM wave level [7]. The complex transmission coefficient matrix for the l -th SIM layer is denoted as $\mathbf{\Gamma}^l = \text{diag}(\mathbf{\Phi}^l)$, where $\mathbf{\Phi}^l = [\phi_1^l, \phi_2^l, \dots, \phi_F^l]^T \in \mathbb{C}^{F \times 1}$, $\phi_j^l = e^{j\kappa_j^l}$, $j = \sqrt{-1}$ and κ_j^l represents the phase shift for each respective layer and meta-atom. Based on the Rayleigh-Sommerfeld diffraction principle [7], the propagation coefficient, which determines the EM wave transmission between the f' -th meta-atom in the $(l-1)$ -th SIM layer and the f -th meta-atom in the l -th layer, is described by

$$[\mathbf{B}^l]_{f,f'} = \frac{d_a s_a}{(d_{f,f'}^l)^2} \left(\frac{1}{2\pi d_{f,f'}^l} - j \frac{1}{\lambda} \right) e^{j2\pi \frac{d_{f,f'}^l}{\lambda}}, \quad \forall l \in \mathcal{L} \forall f, f' \in \mathcal{F}, \quad (1)$$

where λ is the wavelength, $d_{f,f'}^l$ represents the distance between the respective meta-atoms, and s_a is the area of each meta-atom. The inter-layer separation d_a is defined as $\frac{D}{L-1}$, where D indicates the total thickness of the SIM. For the initial layer, $\mathbf{B}^1$ is obtained by replacing $d_{f,f'}^l$ with $d_{f,k}^1$, which denotes the distance between the f -th meta-atom in the first layer and the k -th antenna element of the BS. Thus, the overall EM response of the SIM is represented as

$$\mathbf{\Omega} = \mathbf{\Gamma}^L \mathbf{B}^L \mathbf{\Gamma}^{L-1} \mathbf{B}^{L-1} \dots \mathbf{\Gamma}^2 \mathbf{B}^2 \mathbf{\Gamma}^1 \in \mathbb{C}^{F \times F}. \quad (2)$$

C. Transmission Protocol

To represent the information symbol generated at the BS for each user $u \in \mathcal{U}$, we define $s_u \sim \mathcal{CN}(0, 1)$. The transmitted signal from the BS, therefore, becomes $x = \sum_{u \in \mathcal{U}} \sqrt{p_u} s_u$, where p_u denotes the power allocated to the user symbol s_u . Following the application of wave-domain beamforming at the SIM, the signal received by user u is formulated as $y_u = (\mathbf{h}_u)^T \mathbf{\Omega} \mathbf{b}_u^1 x + z_u$, $\forall u \in \mathcal{U}$. Here, $\mathbf{b}_u^1 \in \mathbb{C}^{F \times 1}$ represents the u -th column of $\mathbf{B}^1$, which characterizes the propagation coefficient between the SIM's first layer and the

u -th antenna element of the BS, associated with user u ¹. The term $z_u \sim \mathcal{CN}(0, \sigma_u^2)$ captures the additive white Gaussian noise (AWGN) at the u -th user, with σ_u^2 as the noise variance. Moreover, $\mathbf{h}_u \in \mathbb{C}^{1 \times F}$ describes the channel characteristics between the SIM and user u , presumed to follow a quasi-static, flat-fading model with full channel state information available to the SIM, based on a standard ray-based model [13]. For each user $u \in \mathcal{U}$, the resulting signal-to-interference-plus-noise ratio (SINR) is computed as

$$\text{SINR}_u = \frac{|\mathbf{h}_u \mathbf{\Omega} \mathbf{b}_u^1|^2 p_u}{\sum_{\substack{u' \in \mathcal{U} \\ u' \neq u}} |\mathbf{h}_u \mathbf{\Omega} \mathbf{b}_{u'}^1|^2 p_{u'} + \sigma_u^2}, \quad \forall u \in \mathcal{U}. \quad (3)$$

Based on the Shannon-Hartley theorem, the achievable data rate for each user u is expressed as

$$R_u = \log_2(1 + \text{SINR}_u), \quad \forall u \in \mathcal{U}. \quad (4)$$

D. QoE Preliminaries

To enable an efficient QoE-oriented resource allocation in wireless networks, as exemplified in [4]–[6], we consider the MOS as the primary QoE metric. This approach relies on diverse services, including web browsing, HTTP live video streaming and VoLTE audio, we discuss the specific MOS model for each, distinctly.

1) *Web Browsing*: For a user u accessing web services, the MOS is formulated as [4] $\text{MOS}_{\ell,u} = -K_1 \ln(\nu(R_u)) + K_2$, where $\ell \in \{\text{Web}\}$ and $\nu(R_u)$ represents the latency experienced by user u , measured from the moment the request is made until the webpage content is fully displayed. This latency is expressed as $\nu(R_u) = 3\text{RTT} + \frac{\text{FS}_u}{R_u} + \left(\frac{\text{MSS}}{R_u} + \text{RTT}\right) \gamma_1 + \frac{2\text{MSS}(2^{\gamma_1} - 1)}{R_u}$, $\forall u \in \mathcal{U}$ where RTT stands for the round-trip time (in seconds), FS_u denotes the file size of the webpage requested by the user u in bits, and MSS is the maximum segment size in bits. Furthermore, γ_1 captures the required cycles for data download, factoring in idle periods during the transmission control protocol (TCP) and HTTP slow-start phases [5]. Specifically, $\gamma_1 = \min\{\gamma_2, \gamma_3\}$, where γ_2 indicates the cycles needed for the congestion window to reach the bandwidth-delay product, and γ_3 reflects the slow-start cycles required to fully exchange the webpage data between the SIM and the user. These values are defined as $\gamma_2 = \left\lceil \log_2 \left(\frac{R_u \text{RTT}}{\text{MSS}} + 1 \right) \right\rceil - 1$, $\forall u \in \mathcal{U}$ and $\gamma_3 = \left\lceil \log_2 \left(\frac{\text{FS}_u}{2\text{MSS}} + 1 \right) \right\rceil - 1$, $\forall u \in \mathcal{U}$.

2) *HTTP Live Video Streaming*: For HTTP-based live video streaming, where H.264/AVC encoding is applied, the MOS for a typical user u can be described as [5]

¹Antenna selection at the BS is excluded here for simplicity, assuming a predefined mapping where each transmit stream directly corresponds to a specific antenna.

$$\text{MOS}_{\ell,u} = \begin{cases} 4.5, & \Xi_u > \Xi^{4.5}, \\ K_3 \log(\Xi_u) + K_4, & \Xi^1 \leq \Xi_u \leq \Xi^{4.5}, \\ 1, & \Xi_u < \Xi^1, \end{cases} \quad \forall u \in \mathcal{U}. \quad (5)$$

Here, $\ell \in \{\text{Video}\}$. Besides, K_3 and K_4 are constants, given by $K_3 = \frac{3.5}{\log(\Xi^{4.5}) - \log(\Xi^1)}$, and $K_4 = \frac{\log(\Xi^{4.5}) - 4.5 \log(\Xi^1)}{\log(\Xi^{4.5}) - \log(\Xi^1)}$, where Ξ_u denotes the peak-signal-to-noise-ratio (PSNR) for user u , calculated as [4] $\Xi_u = K_5 + K_6 \sqrt{\frac{R_u}{K_7}} \left(1 - \frac{K_7}{R_u}\right)$. In this equation, K_5 , K_6 , and K_7 are constants determined by the encoding parameters. Meanwhile, the values Ξ^1 and $\Xi^{4.5}$ represent the minimum acceptable and maximum achievable MOS, respectively [5].

3) *VoLTE Service*: In the realm of audio transmission, and specifically VoLTE services, the R -factor serves as the principal QoE metric, as endorsed by ITU-T Recommendation G.107 [14]. This factor, spanning from 0 to 100, reflects the perceived voice quality, from least to most satisfactory. Specifically, for a typical user u of audio service (i.e., $\ell \in \{\text{Audio}\}$), $\text{MOS}_{\ell,u} = 1$, when $\text{RF}_u < 1$ and $\text{MOS}_{\ell,u}^{\text{Audio}} = 4.5$, when $\text{RF}_u > 1$, $\forall u \in \mathcal{U}$. Otherwise, when $1 < \text{RF}_u < 100$, one can acquire MOS of this user as $\text{MOS}_{\ell,u} = 1 + 0.035 \text{RF}_u + \text{RF}_u (\text{RF}_u - 60) (100 - \text{RF}_u) 7.10^{-6}$, $\forall u \in \mathcal{U}$. Here, RF_u signifies the R -factor for user u . Using the R -factor as a comparative benchmark and guiding parameter, a QoE model for audio services-primarily driven by achievable data rate-can be expressed as $\text{MOS}_{\ell,u} = K_8 \log(R_u) + K_9$, where K_8 and K_9 are constants tailored to the minimum and maximum target rates for user u that ensure satisfactory VoLTE service.

E. Problem Definition

Our objective, here, is to maximize the aggregate achievable QoE of users of different multimedia services. Towards this objective, it becomes essential to co-optimize the transmission power of the BS and EM response matrix of the SIM, i.e., the set of system variables $\Psi = \{\mathbf{P}, \mathbf{\Omega}\}$. Under the consideration of ensuring the MOS requirement of users and the maximum transmit power budget of the BS, the network-wide problem we discussed above is mathematically formulated as

$$\begin{aligned} \mathcal{P}_\ell : \max_{\Psi} \quad & \sum_{u \in \mathcal{U}} \text{MOS}_{\ell,u} \\ \text{C}_1 : \quad & \text{MOS}_{\ell,u} \geq \text{MOS}_\ell^{\min}, \quad \forall u \in \mathcal{U}, \\ \text{C}_2 : \quad & \sum_{u \in \mathcal{U}} p_u \leq P^{\max}, \\ \text{C}_3 : \quad & \mathbf{\Omega} = \mathbf{\Gamma}^L \mathbf{B}^L \mathbf{\Gamma}^{L-1} \mathbf{B}^{L-1} \dots \mathbf{\Gamma}^2 \mathbf{B}^2 \mathbf{\Gamma}^1, \\ \text{C}_4 : \quad & \mathbf{\Phi}^l = [\phi_1^l, \phi_2^l, \dots, \phi_F^l] \quad \forall l \in \mathcal{L}, \\ \text{C}_5 : \quad & \kappa_f^l \in [0, 2\pi), \quad \forall f \in \mathcal{F}, \forall l \in \mathcal{L}. \end{aligned} \quad (6)$$

By introducing $\ell \in \{\text{Web}, \text{Audio}, \text{Video}\}$, (6), in fact, includes 3 distinct problems. In this formulation, constraint C_1 ensures that each user $u \in \mathcal{U}$ meets its required minimum QoE

level, while constraint C_2 imposes a cap on the BS transmission power, ensuring it remains below the maximum allowable threshold $P^{\max}$. The non-convex structure of both constraint C_1 and the objective function presents significant challenges in achieving a global optimum for problem (6). Traditional iterative methods based on convex optimization, such as those in [11], can yield near-optimal solutions but rely heavily on extensive matrix computations, demanding considerable processing power and resulting in slow convergence-rendering them impractical for time-sensitive applications. Moreover, conventional deep reinforcement learning (DRL) methods, like those in [8], face substantial performance degradation in highly dynamic environments, particularly when user mobility is introduced, which impairs their adaptability and effectiveness.

III. SOLUTION STRATEGY

In response to the complexities and challenges outlined, we propose an adaptive and real-time resource allocation framework that harnesses the robust capabilities of meta-enhanced DRL to seamlessly accommodate the dynamic conditions of wireless networks. This innovative methodology integrates meta-learning to dynamically adjust the training parameters of the DRL algorithm, leveraging historical network observations to inform its adjustments. To formalize this approach, problem (6) is thoughtfully redefined within the context of a MDP. A CQL agent is then employed to optimize the optimization variables of the problem. Throughout the agent's training, meta-learning is intricately embedded, continuously refining and recalibrating the agent's parameters, thus enhancing its adaptability and fortifying its ability to generalize across a broader array of diverse and potentially unforeseen tasks, ensuring superior performance in real-time network environments.

A. MDP Transformation of the Problem

We consider an edge node as the CQL agent, responsible for dynamically allocating the resources of the wireless network. This process is formalized as a MDP, which provides an insightful framework for modeling the system's continual variations with following elements.

- **State Space:** At each discrete time step n , the state of the agent, denoted by $\mathbf{s}[n]$, encapsulates the SINR values observed by users from the previous time step, given by $\mathbf{s}[n] = [\text{SINR}_u[n-1]]$. All possible states across all time intervals as well construct the state space $\mathcal{S}$.
- **Action Space:** At time step n , the agent's action is captured by $\mathbf{a}[n] = [\Psi[n]]$, which consists of the optimization variables of (6). The action space $\mathcal{A}$ spans the full range of feasible actions available at each discrete time step.
- **Reward Function:** The transition from state $\mathbf{s}[n-1]$ to state $\mathbf{s}[n]$ is the outcome of executing the action $\mathbf{a}[n]$ at time n . The instantaneous reward for this transition is given by $\text{Rew}[n] = \sum_{u \in \mathcal{U}} \text{MOS}_{\ell,u}[n] - \sum_{i=1}^5 (\chi_i \times \sum_{u \in \mathcal{U}} \text{MOS}_{\ell,u}[n])$, where for each constraint C_i in (6), we set $\chi_i = 1$ if the constraint is satisfied and $\chi_i = 0$ if it is not. This reward formulation ensures a

balanced trade-off between the desired objective and the adherence to the constraints.

B. CQL

As a reinforcement learning algorithm, CQL is developed to ensure robust and conservative optimization in dynamic environments, particularly where exploration is limited or where the consequences of overestimation in value functions can be detrimental [15]. The core of this algorithm lies in a modified objective function that emphasizes conservative Q-value estimates. Let $Q_\theta(\mathbf{s}, \mathbf{a})$ denote the action-value function parameterized by θ , representing the expected cumulative reward of taking action $\mathbf{a}$ in state $\mathbf{s}$ under policy π_θ . On this basis, a function is designed to lower the Q-values for scenarios that are infrequent or less likely according to the policy, thereby minimizing overestimation risk and encouraging safer action choices, formalized as $J(\theta) = \mathbb{E}_{(\mathbf{s}, \mathbf{a}) \sim \mathcal{D}} [Q_\theta(\mathbf{s}, \mathbf{a})] - \alpha \mathbb{E}_{\mathbf{a}' \sim \pi_\theta(\cdot|\mathbf{s})} [Q_\theta(\mathbf{s}, \mathbf{a}')]]$, where α is a tunable parameter that helps prevent high-risk overestimation and adjusts the level of conservatism, balancing the policy's adherence to high-confidence actions in the data distribution $\mathcal{D}$. To solve an optimization problem within a reinforcement learning framework, CQL leverages a dual process of conservative Q-function learning and policy improvement. At each training iteration, the Q-function Q_θ is updated through a modified Bellman loss, which includes a conservatism penalty as $L(\theta) = (\text{Rew}(\mathbf{s}, \mathbf{a}) + \iota \max_{\mathbf{a}'} Q_\theta(\mathbf{s}', \mathbf{a}') - Q_\theta(\mathbf{s}, \mathbf{a}))^2 + \alpha (\mathbb{E}_{\mathbf{a}' \sim \pi_\theta(\cdot|\mathbf{s})} Q_\theta(\mathbf{s}, \mathbf{a}') - Q_\theta(\mathbf{s}, \mathbf{a}))$, where $\mathbf{s}'$ is the next state and ι is the discount factor. The parameter θ of the Q-function are optimized through gradient descent on the conservative objective $J_{\text{CQL}}(\theta)$. The gradient update rule is expressed as $\theta \leftarrow \theta - \zeta \nabla_\theta J(\theta)$, where ζ is the learning rate. This iterative parameter update stabilizes the learning process, systematically refining the Q-values to prioritize actions that align with high-confidence observations in the training data. As a result, achieves a more reliable optimization by reducing reliance on actions that fall outside the empirical data distribution.

C. Meta-Learning

Meta-learning, often described as “*learning to learn*,” enables models to proficiently adjust and generalize across diverse tasks by developing an adaptive initialization that can expedite learning in novel environments [16]. In this scenario, meta-learning is employed to augment both the convergence rate and stability of the CQL algorithm. By integrating meta-learning, the CQL policy network acquires an optimized initialization, facilitating rapid adjustment to varying task-specific demands. This is accomplished through an inner loop focused on task-centric adaptation as well as an outer loop aimed at cross-task generalization.

For individual task optimization, the inner update operates within a framework of tasks T_i , each sampled from a broader

distribution. The policy gradient for each task T_i is computed as

$$\nabla_{\theta} J_{T_i}(\theta) = \mathbb{E}_{\mathbf{s}, \mathbf{a} \sim \pi_{\theta_{\text{old}}}} \left[\frac{\pi_{\theta}(\mathbf{a}|\mathbf{s})}{\pi_{\theta_{\text{old}}}(\mathbf{a}|\mathbf{s})} A^{\pi_{\theta_{\text{old}}}}(\mathbf{s}, \mathbf{a}) \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}|\mathbf{s}) \right] \quad (7)$$

where the gradient derived is applied to update policy parameters for the specific task. This adaptation, which is regulated by a trust region constraint, is represented as $\theta'_{T_i} = \theta - \eta \cdot \nabla_{\theta} J_{T_i}(\theta)$, η with η representing the inner-loop learning rate. This inner loop directs the parameters to maximize performance for the particular task while confining updates to the CQL's designated trust region, thus maintaining model stability.

Once the inner task-based adjustments completed, the outer update compiles these adaptations across multiple tasks to optimize the policy's generalization capability. This meta-level update adjusts the initial policy parameter θ , ensuring the model delivers high performance across a range of tasks following inner-loop adaptation. The meta-objective across tasks is defined as $J_{\text{meta}}(\theta) = \sum_i J_{T_i}(\theta'_{T_i})$, and the parameter θ is updated accordingly as $\theta \leftarrow \theta - \beta \nabla_{\theta} J_{\text{meta}}(\theta)$, where β indicates the meta-learning rate. This outer-loop process aligns the initial parameters with the aim of facilitating rapid convergence to optimal task solutions through subsequent inner-loop refinements, ultimately enhancing learning efficiency across diverse tasks.

D. Complexity Analysis

The computational complexity inherent to DRL-based solutions are primarily determined by the count of layers and neurons within their actor and critic networks, which are fixed parameters and do not scale with network-specific variables such as the number of antennas or users. This renders the proposed resource allocation framework, which utilizes meta-DRL, particularly feasible for practical applications. Specifically, the computational complexity of this approach is split across two main components: the inner-update and outer-update processes. For the inner-update phase, the computational complexity is defined by $\mathcal{O}\left(\sum_{l=0}^{N_L} \Gamma_l \Gamma_{l+1} \times |\mathcal{B}^{\text{inner}}| \times E^{\text{inner}} \times N \times \sum_i T_i\right)$, where N_L denotes the total number of neural network layers, and Γ_l signifies the neuron count in layer l . Additionally, $|\mathcal{B}^{\text{inner}}|$ is the inner-update batch size, E^{inner} the number of episodes within this phase, $\sum_i T_i$ represents the aggregate task count, and N denotes the number of time slots. Likewise, the computational complexity for the outer-update phase can be expressed as $\mathcal{O}\left(\sum_{l=0}^{N_L} \Gamma_l \Gamma_{l+1} \times |\mathcal{B}^{\text{outer}}| \times E^{\text{outer}} \times \sum_i T_i\right)$, where $|\mathcal{B}^{\text{outer}}|$ represents the size of the replay buffer, and E^{outer} denotes the episode count within the outer-update stage. Due to the relatively restrained set of parameters-such as layers, neurons, and tasks-the overall computational complexity remains comparatively low. This is in contrast to traditional convex optimization methods, whose computational complexities often increase exponentially with the dimensions of the network.

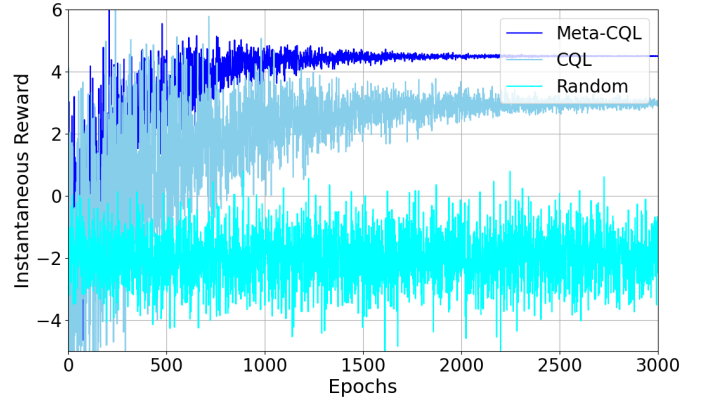


Fig. 2: Convergence behavior of the proposed resource allocation scheme.

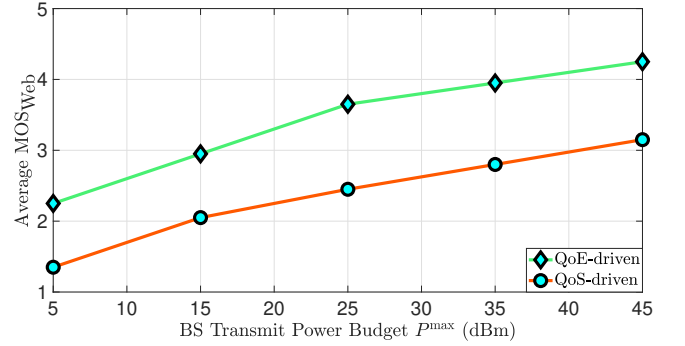


Fig. 3: Average Web MOS vs. the number of SIM layers.

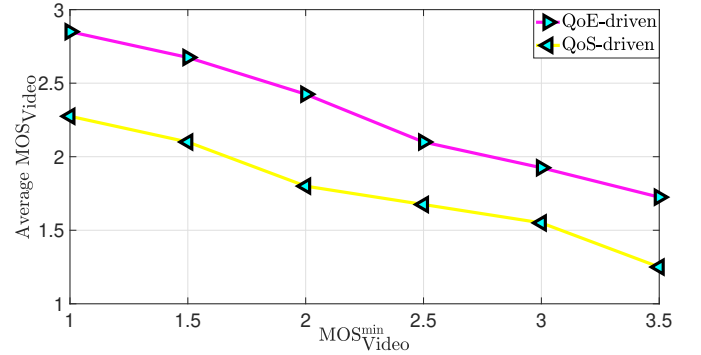


Fig. 4: Average video MOS vs. the BS transmit power budget.

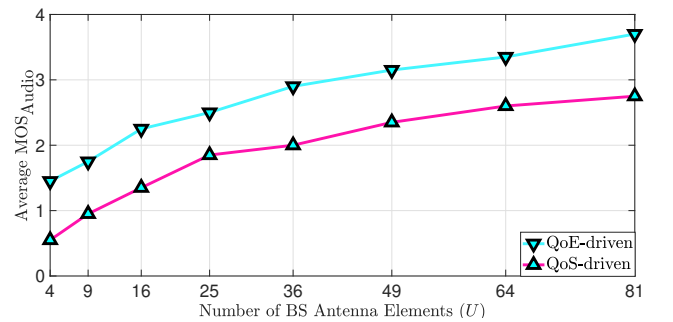


Fig. 5: Average audio MOS vs. the number of BS antenna elements.

IV. PERFORMANCE EVALUATION

This section delves into the evaluation of the proposed network and resource allocation scheme. We have considered $U = 50$, $N = 100$, $\sigma^2 = -174\text{dBm}$, $\tau = 0.1\text{sec}$, $P^{\max} = 46\text{dBm}$, $\beta = \eta = 0.5$, $N_L = 5$, $\sum_i T_i = 100$, $E^{\text{inner}} = E^{\text{outer}} = 100$, $|\mathcal{B}^{\text{inner}}| = |\mathcal{B}^{\text{outer}}| = 10$, $\Gamma_l = 100, \forall l \in N_L$ as simulation parameters, if otherwise stated. Besides, SIM and QoE configurations are respectively derived from [7] and [1], respectively.

Fig. 2 shows the convergence patterns of the proposed resource allocation framework. Meta-CQL demonstrates a significantly faster convergence speed than traditional CQL, largely due to its ability to adapt more readily to the dynamic conditions within the environment. Additionally, this method consistently achieves superior results relative to standard CQL, a benefit that can be attributed to its meta-learning feature, which enhances the algorithm's reactivity. This performance gain is even more pronounced when compared to the random allocation baseline. Notably, the presence of negative values in the illustration arises from the penalty structure of the reward function $Rew(s[n], a[n])$, which applies deductions whenever constraints $C_1 - C_5$ are breached.

Fig. 3 depicts the effect of increasing the BS's transmit power budget on the average web MOS. An upward trend is observed, which aligns with optimization principles. In particular, by permitting higher values for $P^{\max}$ as per constraint C_2 , the feasible solution set for problem (6) becomes less constrained. This expanded feasible region broadens the solution space for Ψ , leading to an enhancement in the achievable average web MOS, aligning with the objective function of the problem. Furthermore, the figure highlights a 31% improvement in average web MOS when utilizing our proposed QoE-focused resource allocation strategy in SIM-enabled systems, compared to traditional QoS-based approaches, such as in [7].

Fig. 4 illustrates the influence of a minimum video MOS requirement on users' average video MOS. The descending trend observed is grounded in principles of optimization theory. In essence, imposing higher values for $MOS_{\text{Video}}^{\min}$ per constraint C_1 restricts the feasible solution set for the optimization problem, thereby reducing the search space for feasible values of Ψ . This restriction leads to a decline in the average video MOS for users, aligning with the objective function in problem (6). Additionally, this figure highlights a notable enhancement (44%) in average video MOS achieved by our proposed QoE-oriented resource allocation approach over conventional QoS-based techniques within SIM-enabled networks, as exemplified by [7].

As illustrated in Fig. 5, an increase in the number of BS antenna elements is associated with a notable rise in the average audio MOS. This relationship highlights the impact of larger BS antenna arrays in enhancing overall system performance. Nonetheless, because we maintain a parity between the number of BS antenna elements and the number of users being served (a one-to-one mapping between the BS antenna elements and the user streams), the trend begins to plateau. This effect is

attributed to escalating inter-user interference as the user count rises, leading to a saturation in the average audio MOS. Notably, this approach achieves approximately 26% improvement in average audio MOS compared to existing QoS-oriented resource allocation methods, such as those outlined in [7].

V. CONCLUSIONS

This research has investigated the efficiency of a SIM-enabled network in handling multimedia applications, including web, video, and audio services. A machine learning-based, QoE-centric resource allocation framework has been crafted to rigorously assess system capabilities, revealing notable performance enhancements over comparable QoS-focused counterparts documented in prior studies.

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